Elliot Miller

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Technical Skills

Languages: C++ (17-23), Rust, C, Assembly (x86_64/ARM), Dafny, Python, C#
Libraries/Frameworks: SDL.h, Tokio, AWS CDK/SDK, Unity
Developer Tools:Git, AWS, AWS Lambdas, Docker, MacOS, Linux, NVim

Highlights

- Was ranked #3 on the Stack Overflow global leaderboards for discussing C++ language semantics (see my website for proof)

Experience

Incoming Software Engineering Intern (Rust) May 2026 – Aug 2026
Apple *Seattle, WA*

- Incoming summer software/systems engineering intern on the Cloud Elastic Disk team, which is developing new storage primitives from the kernel level for the Apple ecosystem and internal ML workloads

Software Engineering Intern (Rust) Jan 2026 – May 2026
N1 (Founder's Fund) *New York, NY*

- Contributing production code to a performant NASDAQ-like central limit order book matching engine handling more than $2B in YTD volume and more than 100k trades per day at a startup backed by Founder's Fund, Kraken, Amber, and Anatoly Yakovenko
- 0 → 1 designed and implemented high-performance rate limiter inspecting every trader action (orders, deposits, etc.) between the NIC and matching engine layer to prevent DOS attacks
- Reduced all database CPU time by 98% (reducing usage from 4 constantly pinned cores to 1 mostly idle core) by indexing expensive queries and editing views to more efficiently join massive tables
- Live-mitigated critical issue halting all deposits and withdrawals

projects

GameBoy Emulator (C++23) Github

- Wrote a fully accurate Z80 Sharp CPU simulator/machine code interpreter capable of running GameBoy ROMs, emulating hardware bugs involving finicky instruction timing and interrupt handling
- Simulated all GameBoy-specific hardware components/systems, including internal clock circuits, interrupts, and proprietary graphics chip

Decentralized Peer to Peer File System (Rust) Github

- Wrote a decentralized P2P file-sharing system in Rust featuring a custom-built DHT based on the Kademlia protocol inspired by the design of BitTorrent.
- Guarded against Sybil and Eclipse attacks by implementing SKademlia security extensions including an upgrade to SHA-2 and POW requirements to join the network

Snake Compiler (Rust, x86)

- Wrote a x86 emitting, optimizing compiler for a Python/OCaml hybrid language called Snake
- Supports Rust interop, register allocation, dead code elimination, and dynamic typing, using lattices to eliminate unnecessary type checks

Multithreading Library (C++20)

- Wrote an efficient implementation of Mutexes and CVs similar to that of the C++ STL in a toy kernel that simulated a multi-core CPU for more realistic concurrent execution

Education

University of Michigan Ann Arbor, MI
Bachelor of Computer Science *Expected May 2027*

- Relevant Coursework: Formal Verification of Systems Software, Distributed Systems, Computer Networks, Operating Systems, Compilers, Web Systems, Computer Organization, Data Structures and Algorithms, Computer Theory, Game Development, Programming Languages